

Understanding the Rendering Equation as a Convolution

1 Introduction

The rendering equation is the foundation of physically based rendering (PBR). It models how light interacts with surfaces and is mathematically expressed as:

$$L_o(\omega_o) = \int_{\Omega} f_r(\omega_o, \omega_i) L_i(\omega_i) (\omega_i \cdot n) d\omega_i. \quad (1)$$

This equation states that the outgoing radiance $L_o(\omega_o)$ at a point is the integral of the incoming radiance $L_i(\omega_i)$ weighted by the surface's Bidirectional Reflectance Distribution Function (BRDF) $f_r(\omega_o, \omega_i)$ and a cosine term $(\omega_i \cdot n)$, which accounts for the angle of incidence.

2 Interpreting the Rendering Equation as a Convolution

Mathematically, convolution is defined as:

$$(f * g)(t) = \int f(\tau) g(t - \tau) d\tau. \quad (2)$$

Comparing this with the rendering equation, we can interpret it as a convolution where:

- $L_i(\omega_i)$ represents the incoming function (irradiance),
- $f_r(\omega_o, \omega_i) \cdot (\omega_i \cdot n)$ acts as the convolution kernel (filter),
- The result is the outgoing radiance $L_o(\omega_o)$, a transformed version of L_i .

Thus, the rendering equation describes how **incoming light is filtered by the surface BRDF**, much like a signal being processed by a convolution filter.

3 Importance Sampling and PDFs in Monte Carlo Integration

Since direct computation of the integral is intractable, Monte Carlo integration is used:

$$L_o(\omega_o) \approx \frac{1}{N} \sum_{i=1}^N \frac{f_r(\omega_o, \omega_i) L_i(\omega_i) (\omega_i \cdot n)}{p(\omega_i)}. \quad (3)$$

The Probability Density Function (PDF), $p(\omega_i)$, determines which directions are sampled more frequently. By choosing $p(\omega_i)$ proportional to $\cos \theta$, we optimize the sampling efficiency, reducing noise while maintaining accuracy.

4 Conclusion

This interpretation lays the foundation for understanding **BxDFs** as convolution filters that shape light reflections, and **Image-Based Lighting (IBL)** as an extension of the rendering equation. The next sections will explore these concepts in detail.